

Meta-analysis with R

G0B75A, Meta-analysis

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An example in education

(Raudenbush, S. W. (1984). Magnitude of teacher expectancy effects on pupil IQ as a function of the credibility of expectancy induction: A synthesis of findings from 18 experiments. *Journal of Educational Psychology*, 76, 85-97.)

Study	Weeks prior contact	g_j	$\sigma_{(g_j)}$
1. Rosenthal et al. (1974)	2	0.03	0.13
2. Conn et al. (1968)	3	0.12	0.15
3. Jose & Cody (1971)	3	-0.14	0.17
4. Pellegrini & Hicks (1972)	0	1.18	0.37
5. Pellegrini & Hicks (1972)	0	0.26	0.37
6. Evans & Rosenthal (1969)	3	-0.06	0.10
7. Fielder et al. (1971)	3	-0.02	0.10
8. Claiborn (1969)	3	-0.32	0.22
9. Kester & Letchworth (1972)	0	0.27	0.16
10. Maxwell (1970)	1	0.80	0.25
11. Carter (1970)	0	0.54	0.30
12. Flowers (1966)	0	0.18	0.22
13. Keshock (1970)	1	-0.02	0.29
14. Henrickson (1970)	2	0.23	0.29
15. Fine (1972)	3	-0.18	0.16
16. Greiger (1970)	3	-0.06	0.17
17. Rosenthal & Jacobson	1	0.30	0.14
18. (1968)	2	0.07	0.09
19. Fleming & Anttonen (1971)	3	-0.07	0.17



Install R & and R studio on your computer.

You can download these from

- o <https://cran.r-project.org/>
- o <https://www.rstudio.com/products/rstudio/download/>



Open the R-script (Toledo: Course Documents > Software > R)



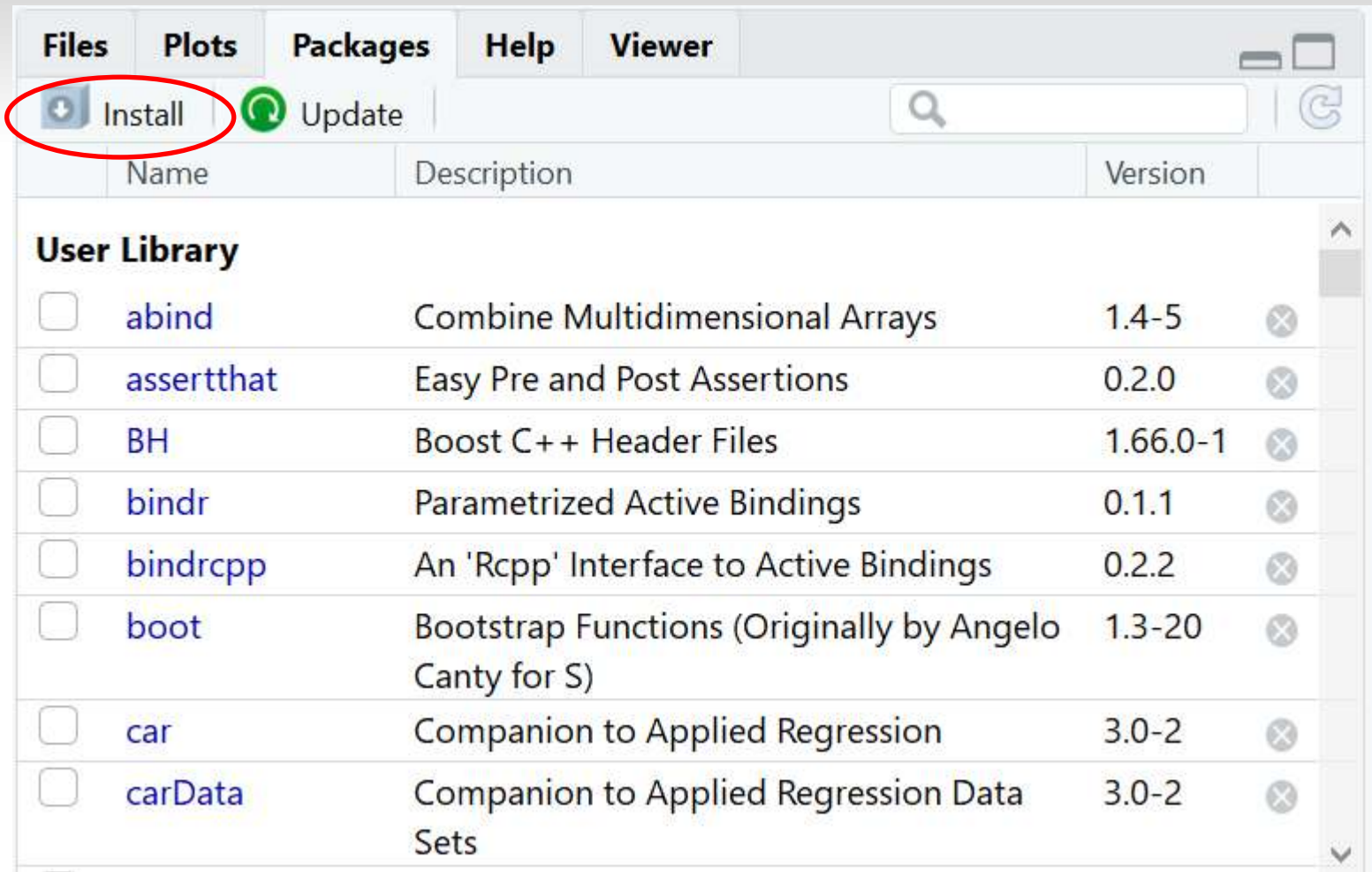
R

Attached Files:

-  ABUSE.xlsx ▾ (37.812 KB)
-  R code Raudenbush example with output.doc ▾ (137 KB)
-  R script ES.R ▾ (1.271 KB)
-  R script MA of diagnostic studies.R ▾ (945 B)
-  R script network meta-analysis.R ▾ (2.971 KB)
-  R studio script multilevel.R ▾ (773 B)
-  R studio script Raudenbush.R ▾ (3.268 KB)
-  Raudenbush.xlsx ▾ (15.926 KB)
-  Reference manual Metafor.pdf ▾ (858.816 KB)



Install the package Metafor

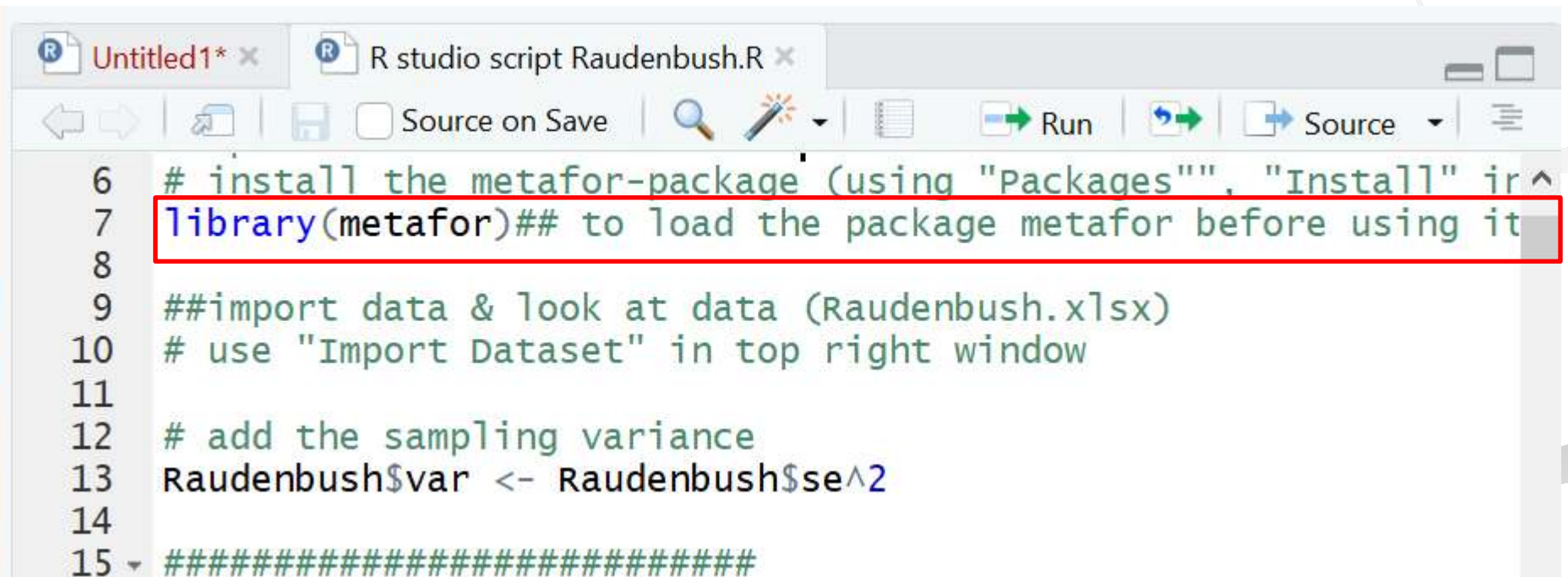


The screenshot shows the RStudio interface with the 'Packages' tab selected. The 'Install' button, represented by a blue square with a white downward arrow, is circled in red. To its right is an 'Update' button with a green circular arrow icon. A search bar is located to the right of the buttons. Below the buttons is a table listing packages in the 'User Library'.

	Name	Description	Version	
User Library				
☐	abind	Combine Multidimensional Arrays	1.4-5	×
☐	assertthat	Easy Pre and Post Assertions	0.2.0	×
☐	BH	Boost C++ Header Files	1.66.0-1	×
☐	bindr	Parametrized Active Bindings	0.1.1	×
☐	bindrcpp	An 'Rcpp' Interface to Active Bindings	0.2.2	×
☐	boot	Bootstrap Functions (Originally by Angelo Canty for S)	1.3-20	×
☐	car	Companion to Applied Regression	3.0-2	×
☐	carData	Companion to Applied Regression Data Sets	3.0-2	×



Load the package Metafor



The image shows a screenshot of the R Studio script editor. The title bar indicates the script is named 'R studio script Raudenbush.R'. The editor contains R code for installing and loading the 'metafor' package. Line 7, which contains the command to load the package, is highlighted with a red rectangular box. The code is as follows:

```
6 # install the metafor-package (using "Packages", "Install" in  
7 library(metafor)## to load the package metafor before using it  
8  
9 ##import data & look at data (Raudenbush.xlsx)  
10 # use "Import Dataset" in top right window  
11  
12 # add the sampling variance  
13 Raudenbush$var <- Raudenbush$se^2  
14  
15 #####
```



Import the dataset (Toledo: Course Documents > Software > R)

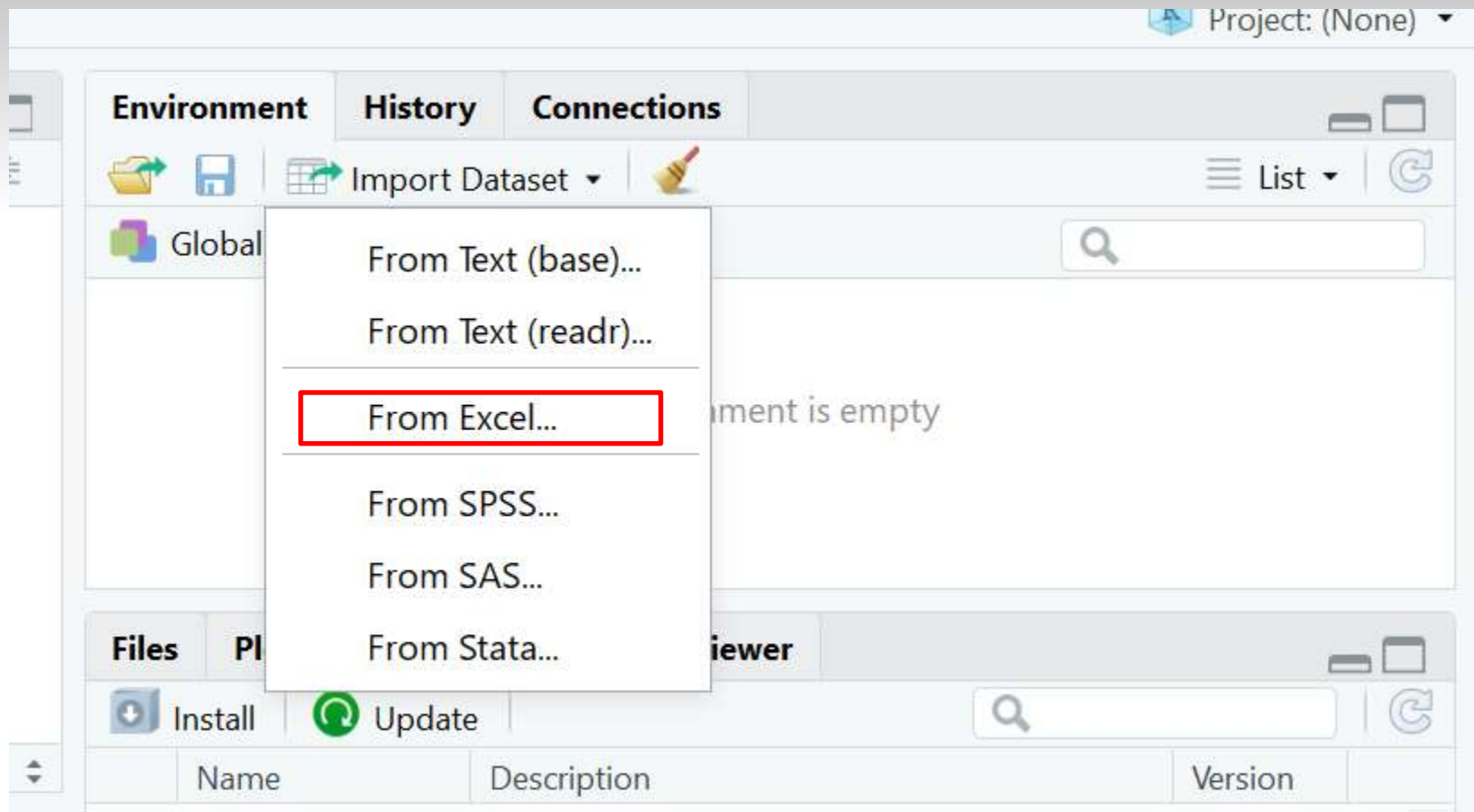


R

- Attached Files:
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Import the dataset



Untitled1* x

R studio script Raudenbush.R x

Raudenbush x

Filter

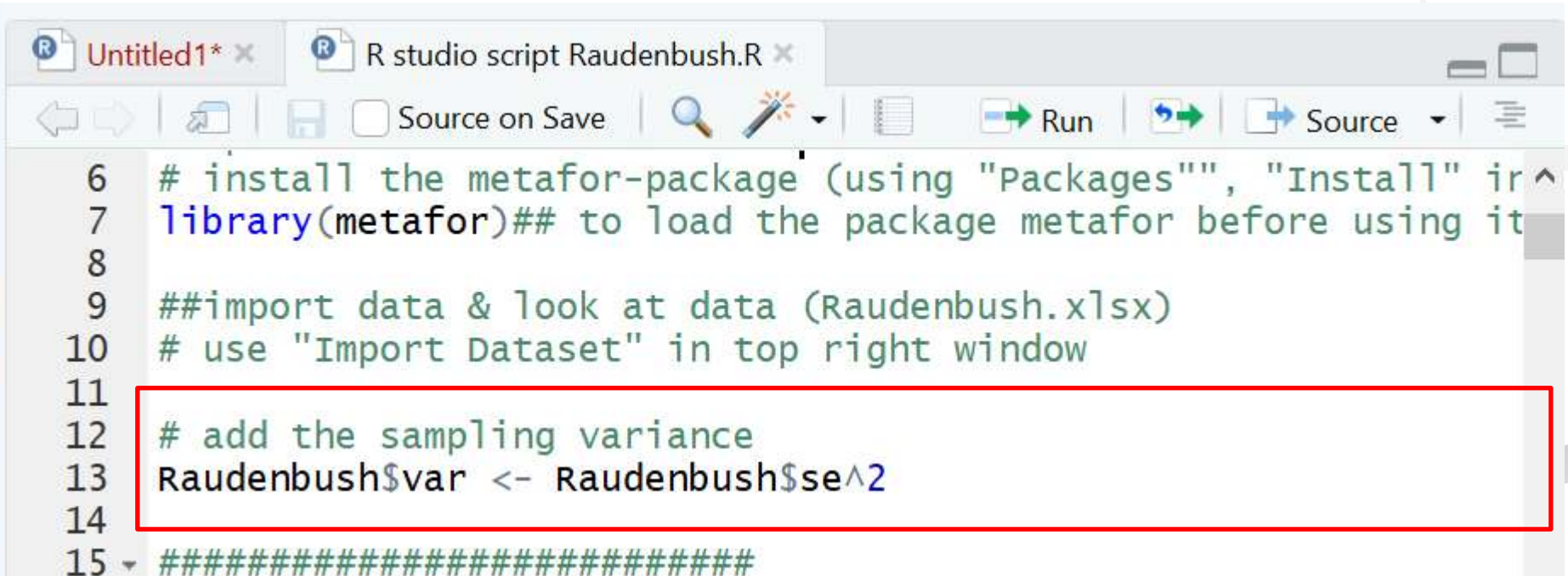


	study	weeks	g	se
1	1	2	0.03	0.125
2	2	3	0.12	0.147
3	3	3	-0.14	0.167
4	4	0	1.18	0.373
5	5	0	0.26	0.369
6	6	3	-0.06	0.103

Showing 1 to 7 of 19 entries



Make a new variable with sampling variances



```
6 # install the metafor-package (using "Packages", "Install" in
7 library(metafor)## to load the package metafor before using it
8
9 ##import data & look at data (Raudenbush.xlsx)
10 # use "Import Dataset" in top right window
11
12 # add the sampling variance
13 Raudenbush$var <- Raudenbush$se^2
14
15 #####
```

The screenshot shows the R Studio interface with a script titled 'R studio script Raudenbush.R'. The code in the editor includes comments and R commands for installing the 'metafor' package, loading it, importing data from 'Raudenbush.xlsx', and calculating a new variable 'var' as the square of the standard error 'se'. The line `Raudenbush$var <- Raudenbush$se^2` is highlighted with a red rectangle.

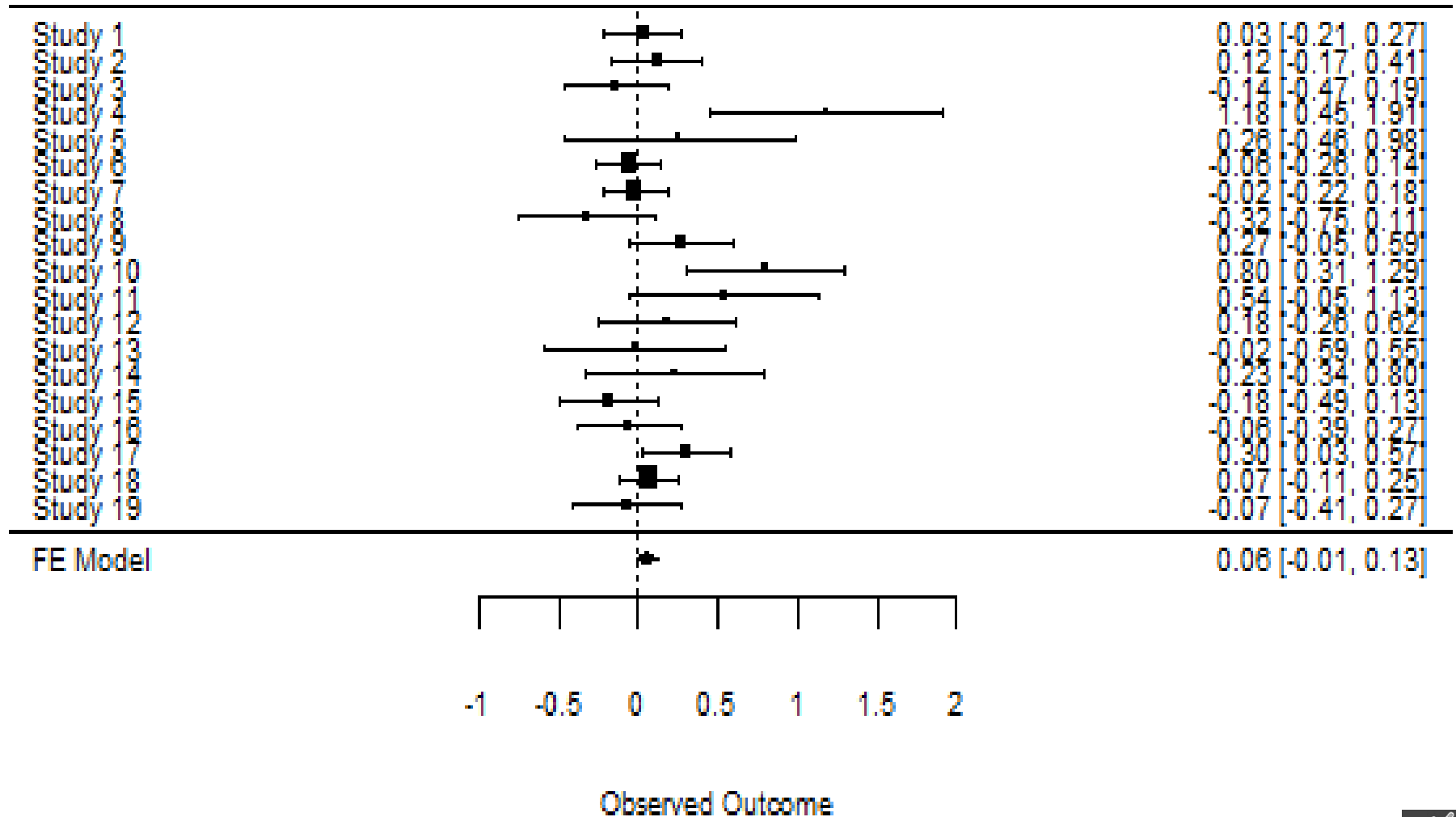
	study	weeks	g	se	var
1	1	2	0.03	0.125	0.015625
2	2	3	0.12	0.147	0.021609
3	3	3	-0.14	0.167	0.027889
4	4	0	1.18	0.373	0.139129
5	5	0	0.26	0.369	0.136161
6	6	3	-0.06	0.103	0.010609

```
FEM<-rma(g, var, data=Raudenbush,  
method="FE")  
summary(FEM)
```

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub
0.0603	0.0365	1.6539	0.0981	-0.0112	0.1319

forest(FEM)



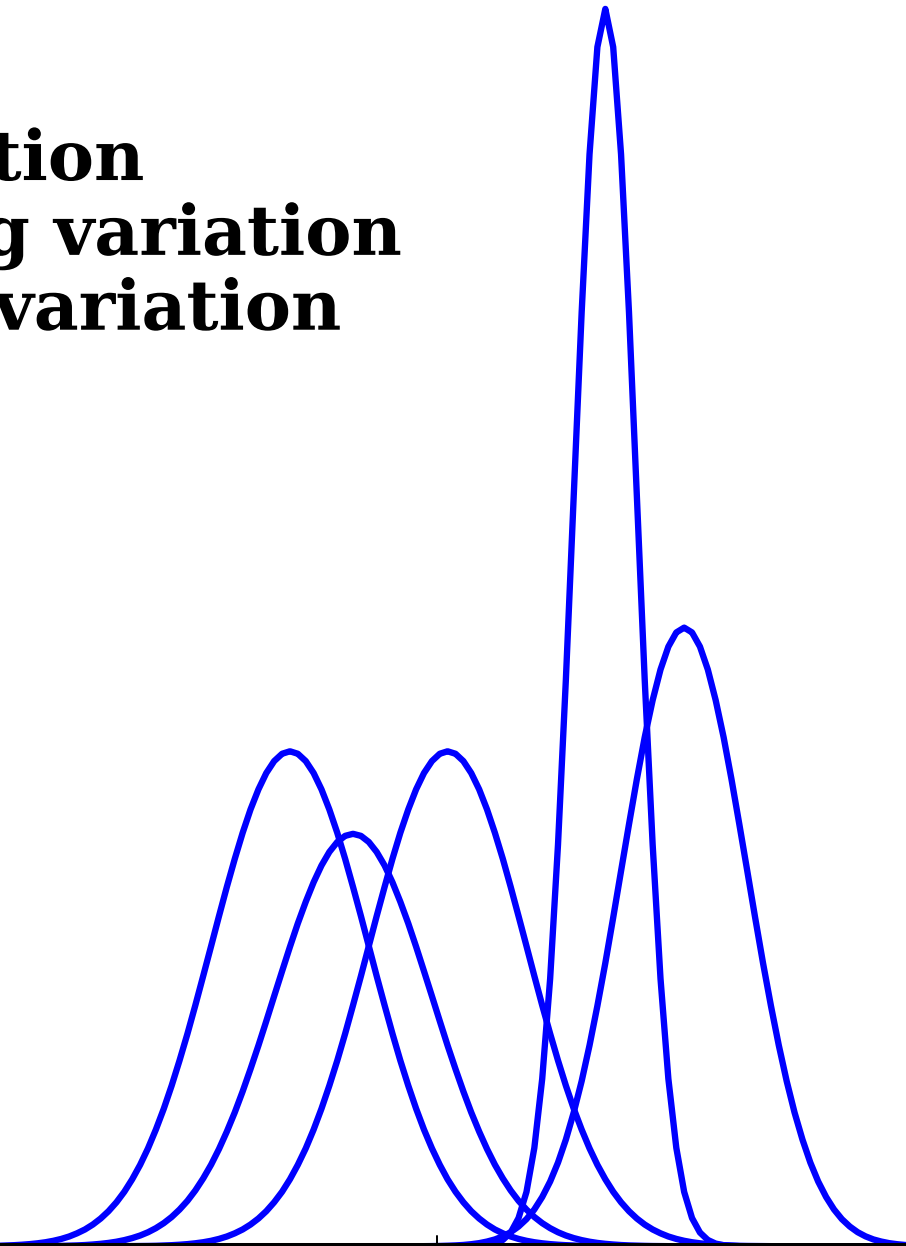
But ...

... studies often differ in:

- Kind of participants (e.g., age)
- Study context
- Outcome variable and its operationalization
- Independent variable and its operationalization
- Study design
- ...



**Total variation
= Sampling variation
+ 'true' variation**



δ_5

δ_1

δ_2

δ_4

δ_3

g



10

Homogeneity?

$$H_0: \delta_1 = \delta_2 = \dots = \delta_k$$

H_a : at least one δ_j differs from an other

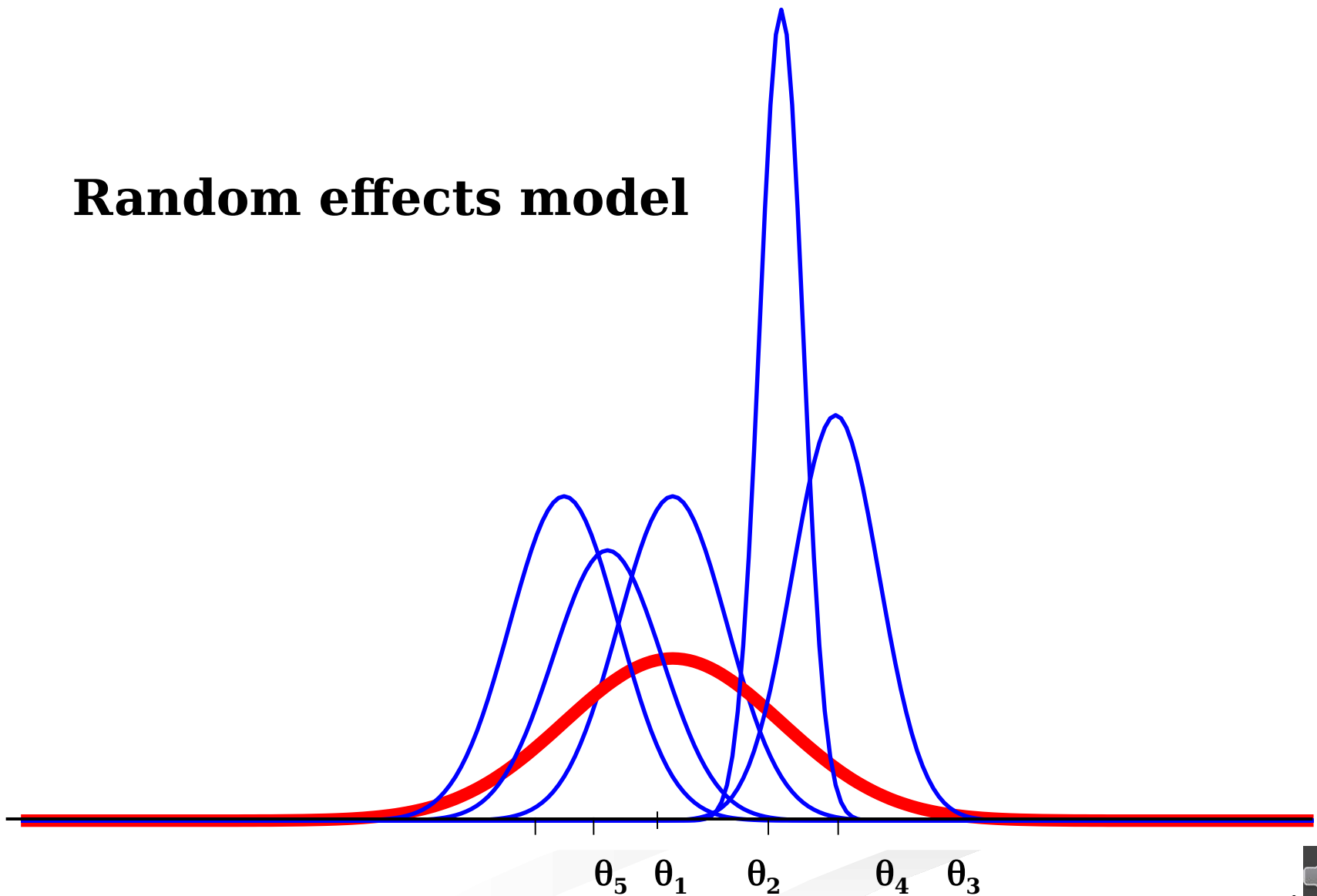
$$Q = \sum_{j=1}^k w_j (g_j - \hat{\delta})^2$$

$$H_0 : Q \sim \chi_{(k-1)}^2$$

Test for Heterogeneity:
 $Q(df = 18) = 35.8254, p\text{-val} = 0.0074$



Random effects model



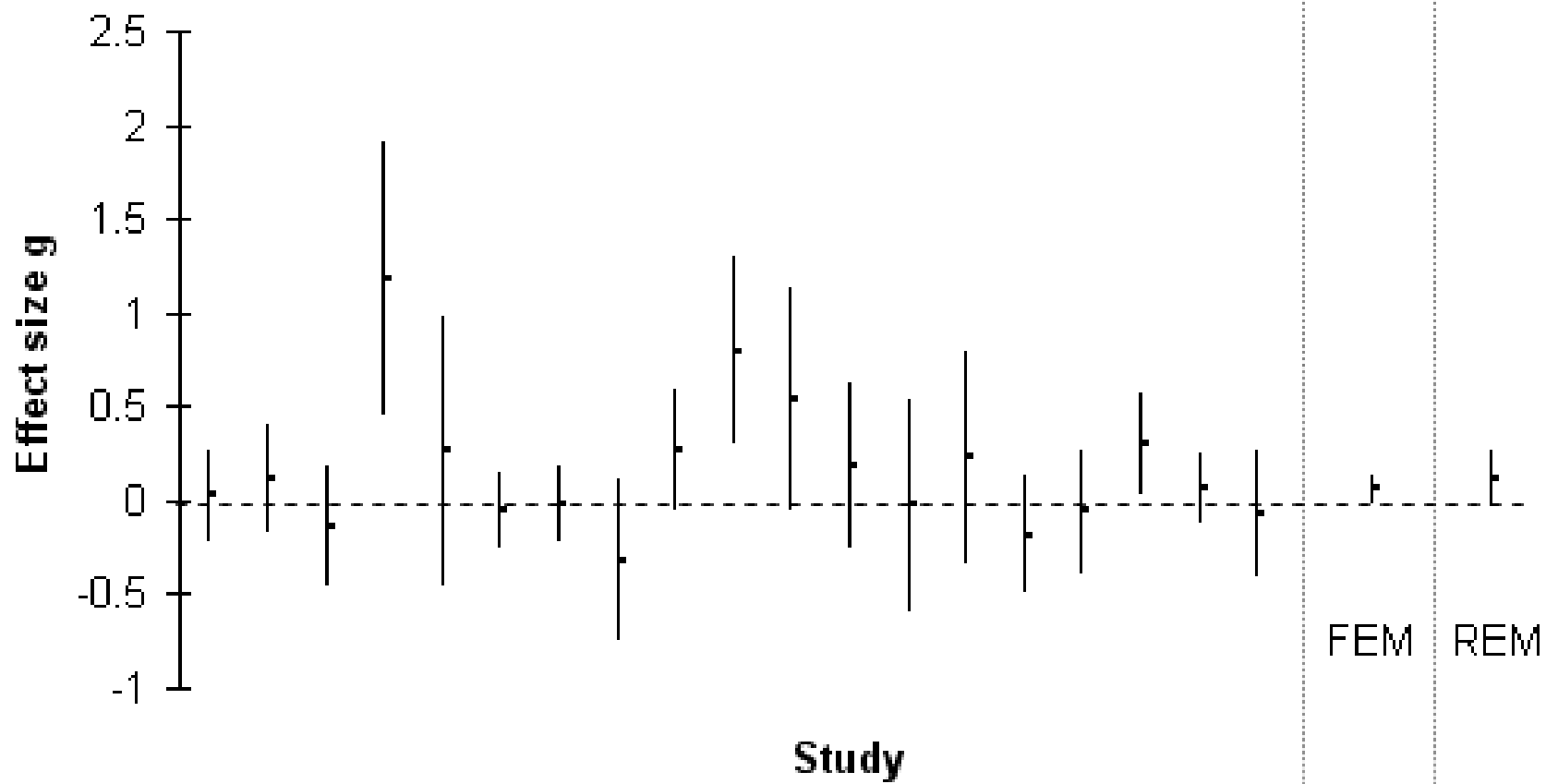
```
REM <- rma(g, var, data=Raudenbush,  
method="REML")  
summary(REM)
```

```
tau^2 (estimated amount of total heterogeneity): 0.0188 (SE = 0.0155)  
tau (square root of estimated tau^2 value):      0.1372  
I^2 (total heterogeneity / total variability):    41.85%  
H^2 (total variability / sampling variability):    1.72
```

```
Test for Heterogeneity:  
Q(df = 18) = 35.8254, p-val = 0.0074
```

```
Model Results:
```

estimate	se	zval	pval	ci.lb	ci.ub
0.0837	0.0517	1.6203	0.1052	-0.0175	0.1849

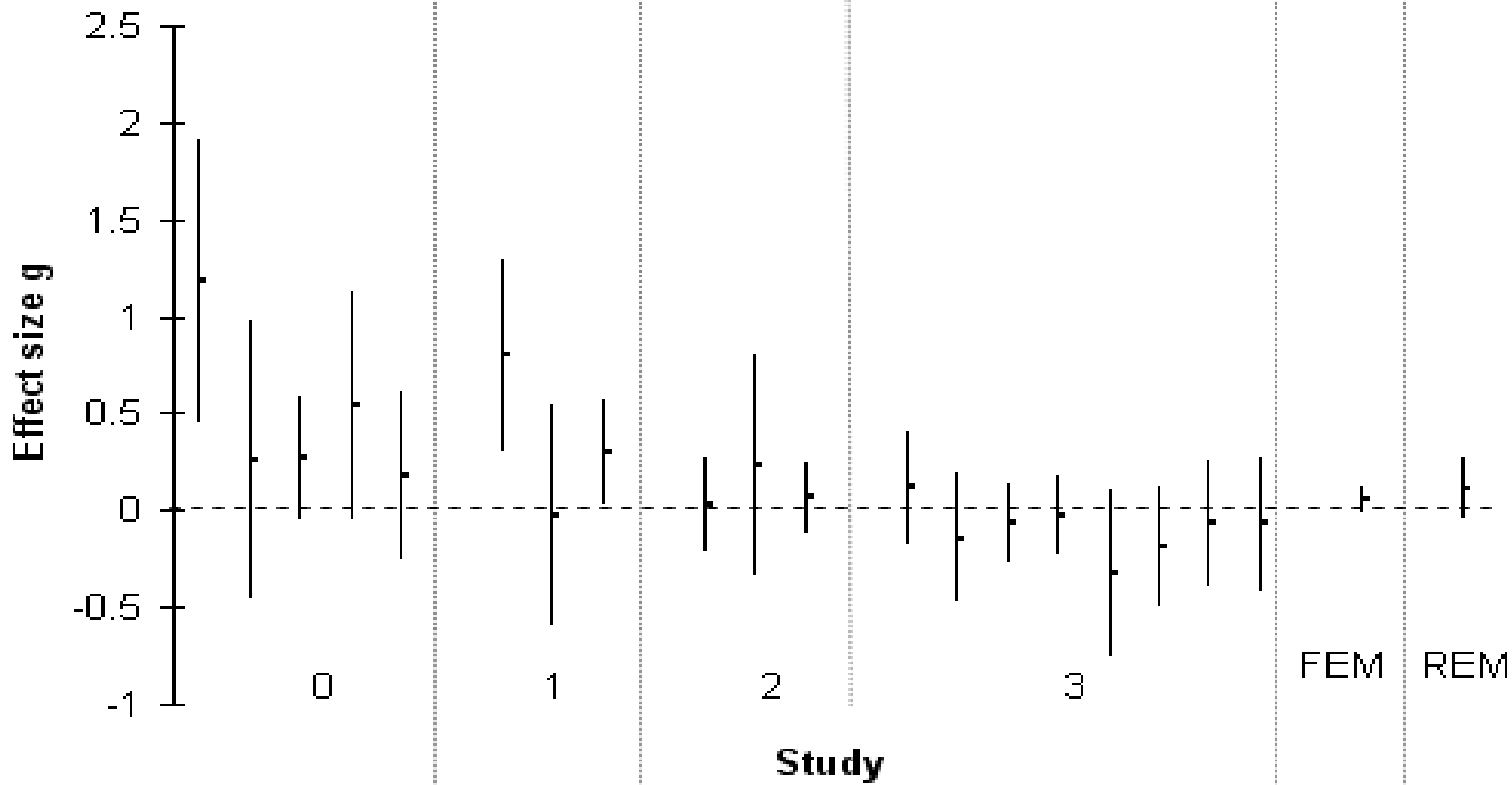


An example in education

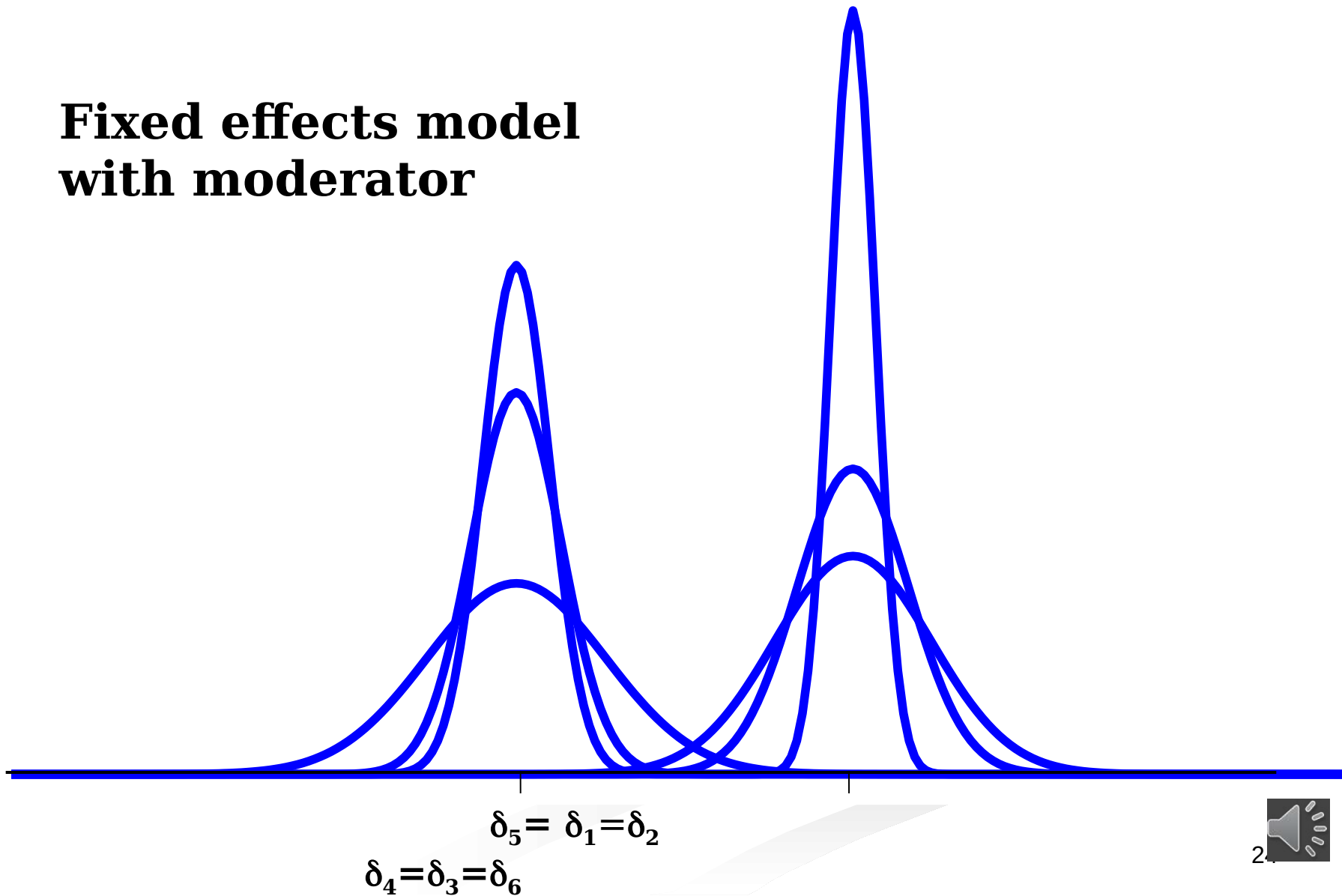
(Raudenbush, S. W. (1984). Magnitude of teacher expectancy effects on pupil IQ as a function of the credibility of expectancy induction: A synthesis of findings from 18 experiments. *Journal of Educational Psychology*, 76, 85-97.)

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11. Carter (1970)	0	0.54	0.30
12. Flowers (1966)	0	0.18	0.22
13. Keshock (1970)	1	-0.02	0.29
14. Henrickson (1970)	2	0.23	0.29
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18. (1968)	2	0.07	0.09
19. Fleming & Anttonen (1971)	3	-0.07	0.17

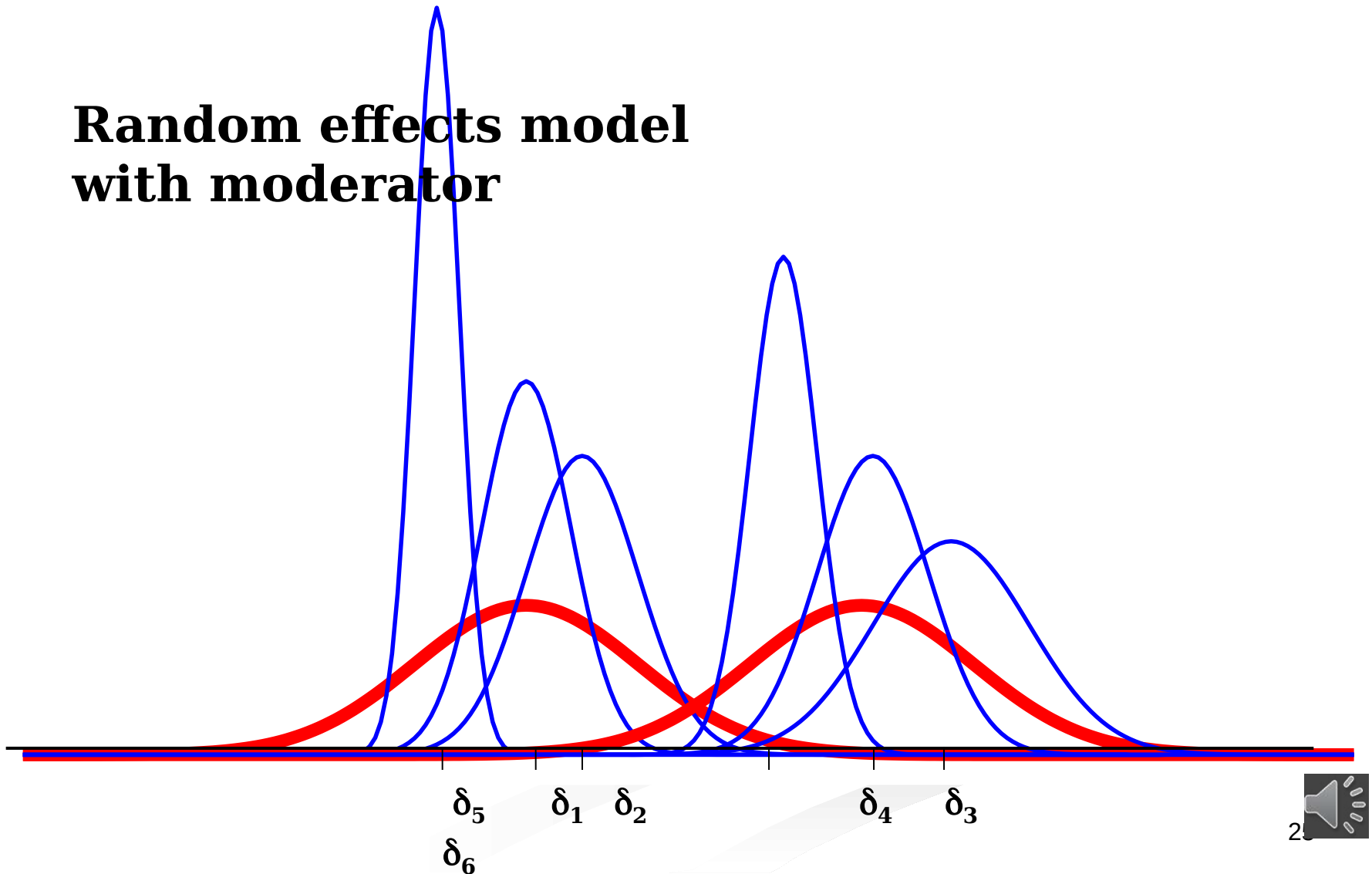




Fixed effects model with moderator



Random effects model with moderator



```
FEMANOVA <- rma(g, var, data = Raudenbush,
  mods = ~ factor(weeks), method = "FE")
summary(FEMANOVA)
```

Test for Residual Heterogeneity:
QE(df = 15) = 15.4466, p-val = 0.4198

Test of Moderators (coefficient(s) 2:4):
QM(df = 3) = 20.3788, p-val = 0.0001

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub	
intrcpt	0.3620	0.1099	3.2939	0.0010	0.1466	0.5774	***
factor(weeks)1	-0.0105	0.1570	-0.0666	0.9469	-0.3181	0.2972	
factor(weeks)2	-0.2955	0.1318	-2.2422	0.0249	-0.5538	-0.0372	*
factor(weeks)3	-0.4251	0.1208	-3.5201	0.0004	-0.6617	-0.1884	***



```
REMANOVA<-rma(g, var, data=Raudenbush,  
mods=~factor(weeks), method="REML")  
summary(REMANOVA)
```

```
tau^2 (estimated amount of residual heterogeneity):      0.0000 (SE = 0.0083)  
tau (square root of estimated tau^2 value):            0.0006  
I^2 (residual heterogeneity / unaccounted variability): 0.00%  
H^2 (unaccounted variability / sampling variability):    1.00  
R^2 (amount of heterogeneity accounted for):            100.00%
```

```
Test for Residual Heterogeneity:  
QE(df = 15) = 15.4466, p-val = 0.4198
```

```
Test of Moderators (coefficient(s) 2:4):  
QM(df = 3) = 20.3785, p-val = 0.0001
```

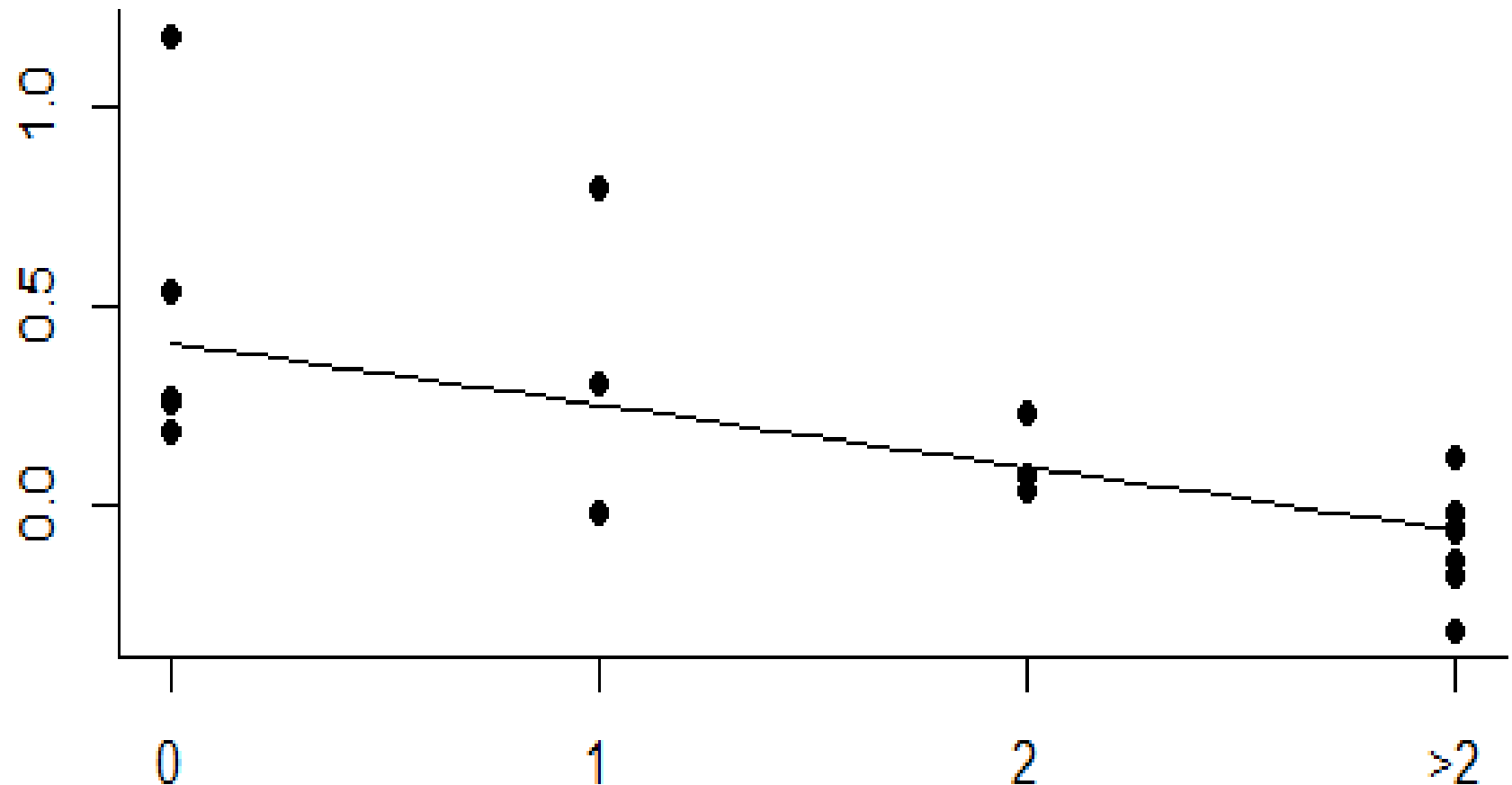
Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub	
intrcpt	0.3620	0.1099	3.2939	0.0010	0.1466	0.5774	***
factor(weeks)1	-0.0105	0.1570	-0.0666	0.9469	-0.3181	0.2972	
factor(weeks)2	-0.2955	0.1318	-2.2422	0.0250	-0.5538	-0.0372	
factor(weeks)3	-0.4251	0.1208	-3.5201	0.0004	-0.6617	-0.1884	***



Fixed/Random effects regression model

Standardized Mean Difference



Weeks of Prior Contact



```
FERM <- rma(g, var, data = Raudenbush,  
mods = weeks, method = "FE")  
summary(FERM)
```

Test for Residual Heterogeneity:
QE(df = 17) = 16.5680, p-val = 0.4840

Test of Moderators (coefficient(s) 2):
QM(df = 1) = 19.2573, p-val < .0001

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	0.4072	0.0871	4.6775	<.0001	0.2366	0.5779
mods	-0.1573	0.0358	-4.3883	<.0001	-0.2275	-0.0870



```
RERM<-rma(g, var, data=Raudenbush,  
mods=weeks, method="REML")  
summary(RERM)
```

```
tau^2 (estimated amount of residual heterogeneity):      0.0000 (SE = 0.0072)  
tau (square root of estimated tau^2 value):             0.0014  
I^2 (residual heterogeneity / unaccounted variability): 0.01%  
H^2 (unaccounted variability / sampling variability):    1.00  
R^2 (amount of heterogeneity accounted for):             99.99%
```

```
Test for Residual Heterogeneity:  
QE(df = 17) = 16.5680, p-val = 0.4840
```

```
Test of Moderators (coefficient(s) 2):  
QM(df = 1) = 19.2562, p-val < .0001
```

```
Model Results:
```

	estimate	se	zval	pval	ci.lb	ci.ub	
intrcpt	0.4072	0.0871	4.6774	<.0001	0.2366	0.5779	***
mods	-0.1573	0.0358	-4.3882	<.0001	-0.2275	-0.0870	***

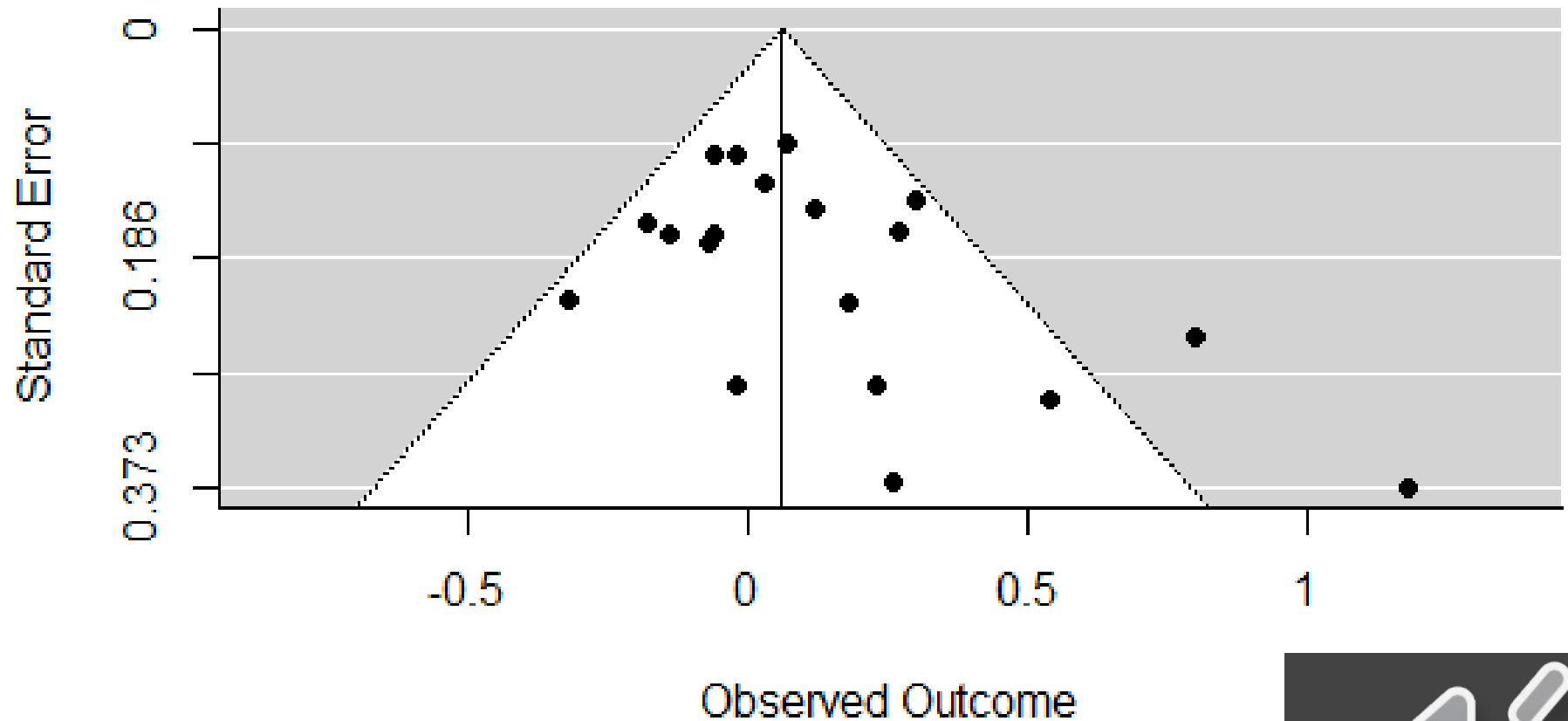


	FEM	REM
Without moderator		
Categorical moderator		
Continuous moderator		

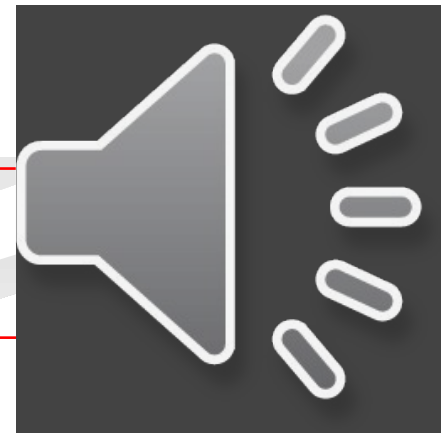
Basic meta-analytic questions

1. Is there an overall effect?
2. How large is this effect?
3. Is the effect the same in all studies?
4. How large is the variation over studies?
5. Is this variation related to study characteristics?

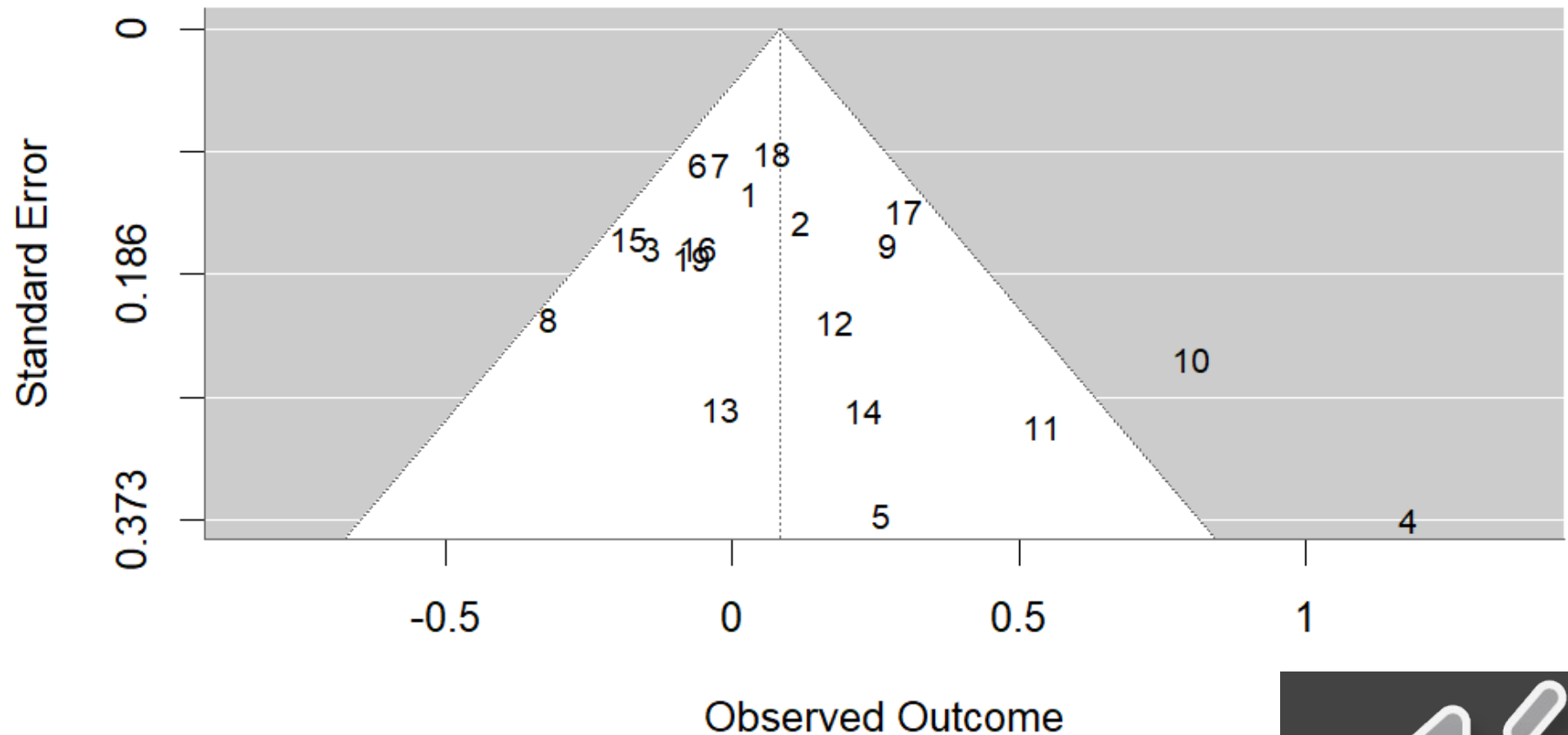
Remark: publication bias!



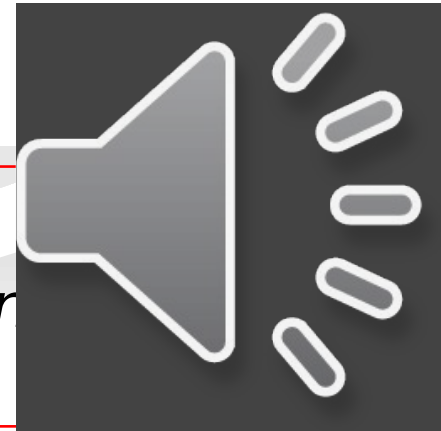
funnel(REM)



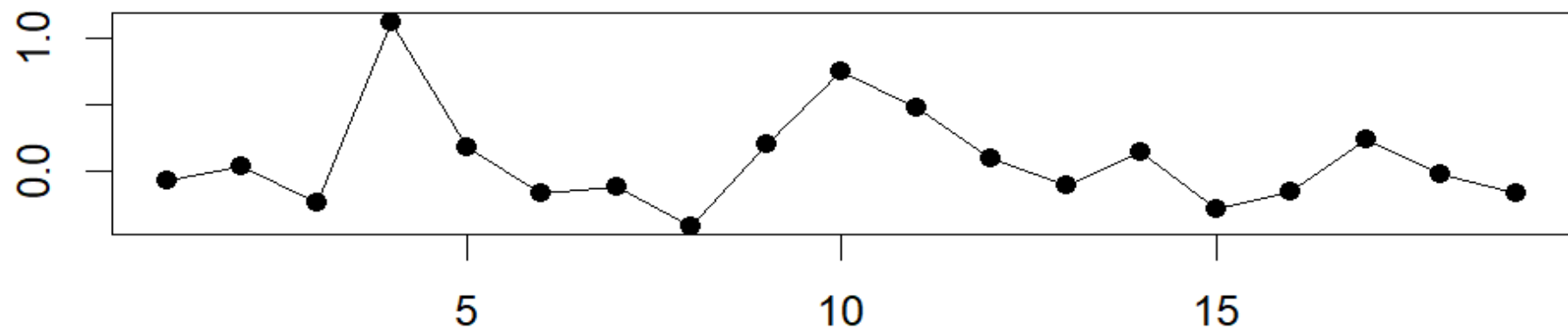
Outlying studies?



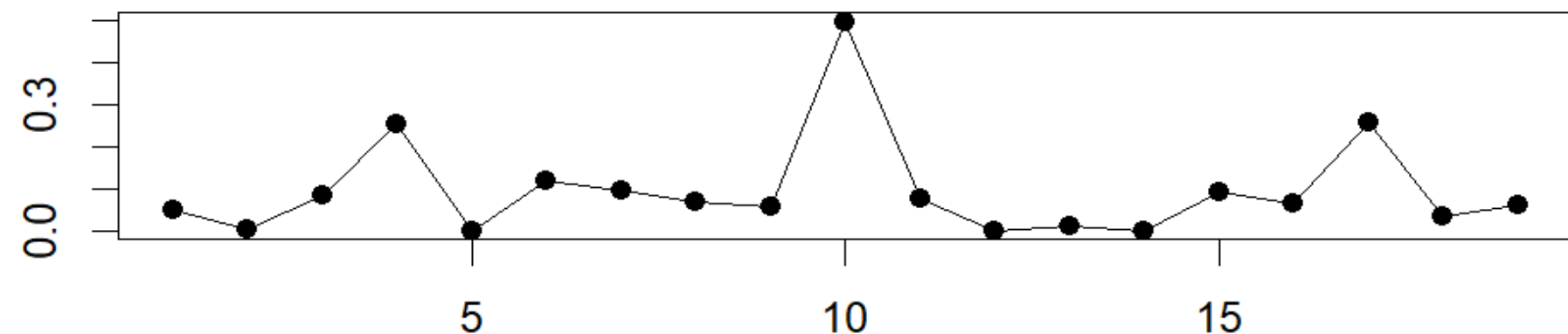
```
funnel(REM, pch=NA)  
text( REM$yi, Raudenbush$se, labels = Raudenbush$label,  
cex=0.8)
```



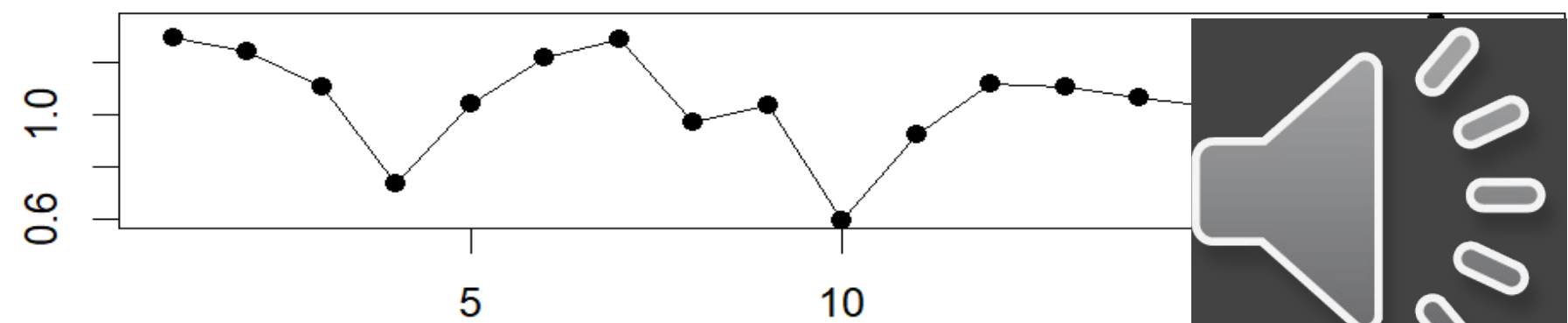
Studentize residuals



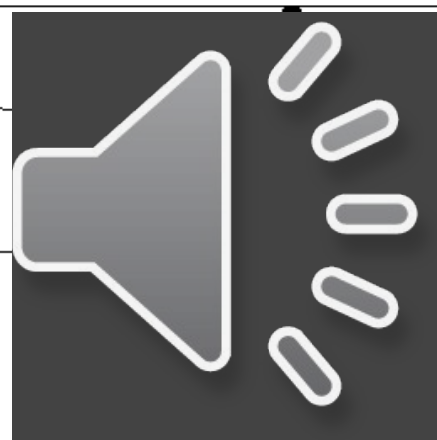
Cook's Distance



Covariance ratio

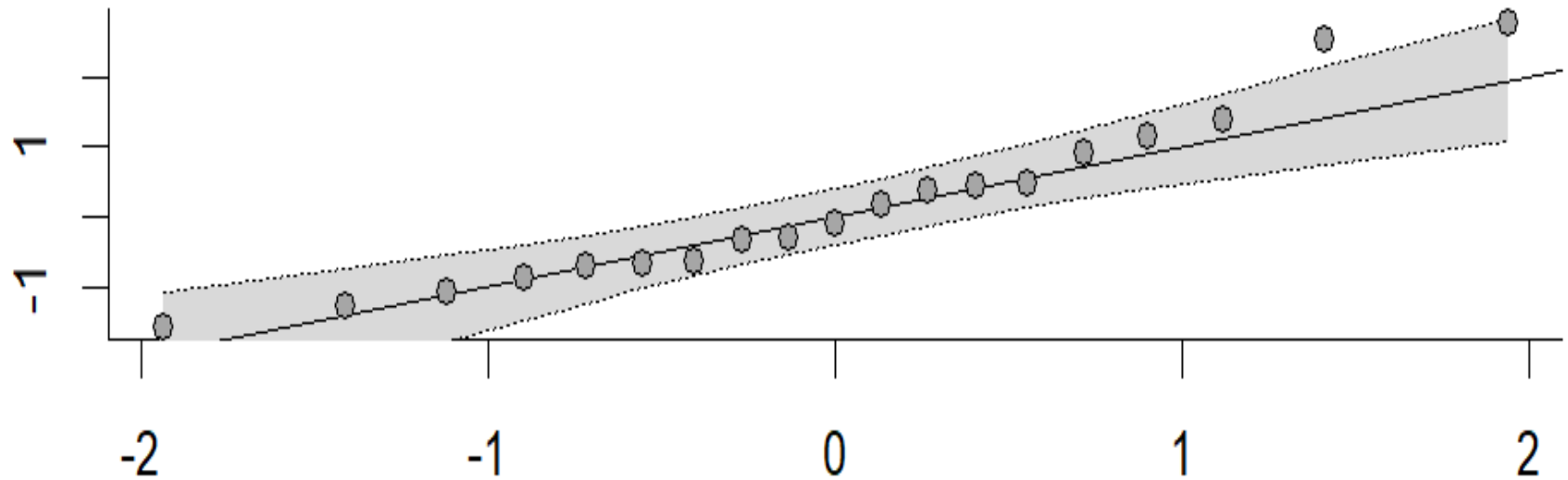


Observed Outcome



Normal Q-Q Plot

Sample Quantiles



Theoretical Quantiles

`qqnorm(REM)`

